

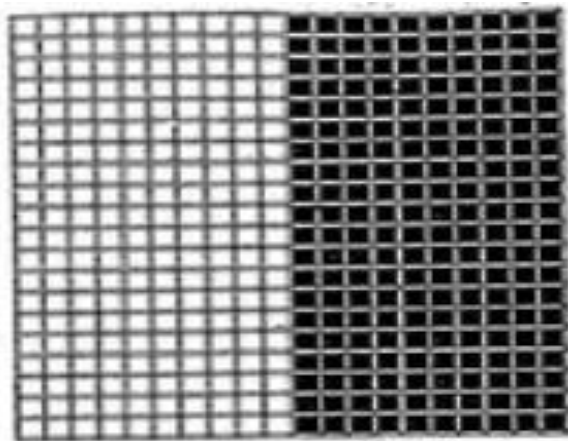
MIXING

Mixing

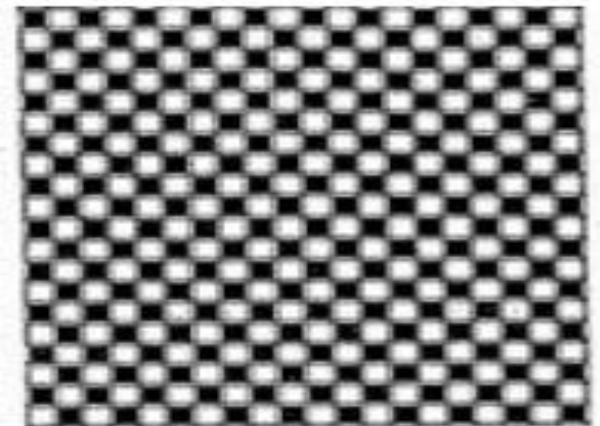
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- Mixing is defined as a process that tends to result in a randomization of dissimilar particles within a system.
- Mixing is a process in which two or more than two components are treated in such a way so that each particle of any one ingredient lies as nearly as possible to the adjacent particles.

Before mixing



After mixing



Factors Affecting Mixing

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- There are factors which affect mixing, these include:
 - Material density
 - Particle size
 - Particle shape
 - Particle attraction
 - Proportion of materials
 - Condition for mixing
 - Mixer volume
 - Mixing mechanism
 - Mixing time
 - Handling of the mixed powder

Liquid mixing

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- Liquid mixing is divided into:

A) Mixing of liquids and liquids

- ***Two miscible liquids***

- Mixing of two miscible liquids is easy and occurs by diffusion
- It can be achieved by simple shaking or stirring

- ***Two immiscible liquids***

- Mixing of two immiscible liquids to form emulsion

Liquid mixing...

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B) Mixing of liquids and solids

□ *Soluble solids*

- Mixture of liquids and soluble solids can occur by simple agitation or stirring
- It can be achieved by simple shaking or stirring

□ *Insoluble solids*

- Mixtures of liquids and insoluble solids requires continuous agitation

Liquid mixing...

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- In liquid mixing, mixing occurs in two stages:

1. localized mixing in which shear is applied to the particles of the fluid
2. a general movement to produce uniform product

Liquid Mixers

- Equipments for liquid mixing include:

- ***Shaker mixer***

- Simplest form of mixer
- Material is agitated by an oscillatory or rotatory movement

Liquid mixing...

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- The efficiency of mixing depends on the degree of turbulence.
- This is also affected by liquid properties.



Industrial shaker mixer



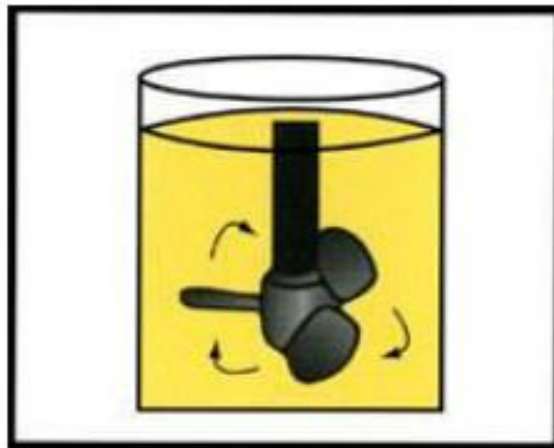
Laboratory shaker mixer

Liquid mixing...

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□ **Propeller mixer**

- Widely used for mixing liquids of low viscosity
- Mixing is achieved through propeller which rotates at a very high speed (i.e. 8000 rpm) due to which mixing occurs in short time.



Propeller blade

Liquid mixing...

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□ **Uses of Propeller mixer**

- Are used when high mixing capacity is needed
- They are effective in handling liquids having a maximum viscosity of about 2 Pas and slurry up to 10% solids of fine mesh size
- Multivitamin elixir and disinfectant solutions are manufactured using propellers

□ **Demerits**

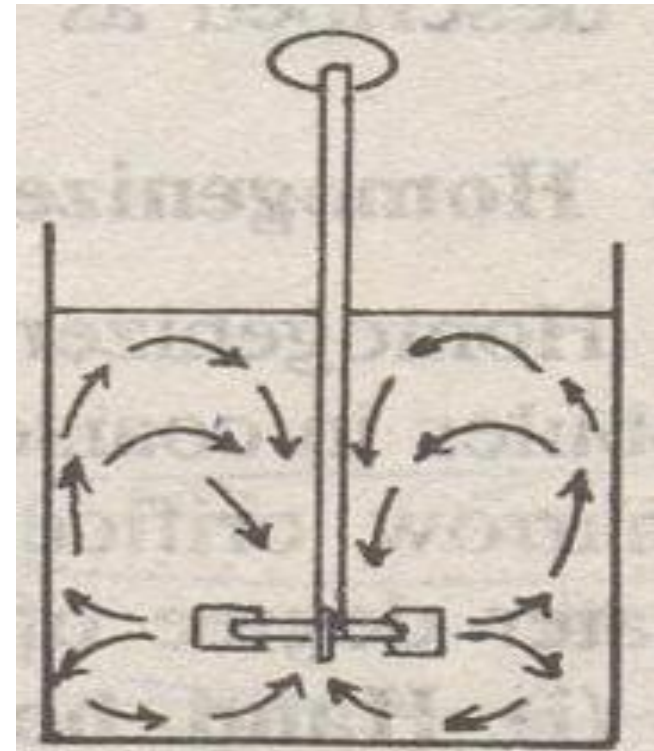
- Are not normally effective with high viscosity liquids

Liquid mixing...

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□ *Turbine mixers*

- It generally consists of circular disc impeller to which a number of short, straight or curved blades are attached.
- **They rotate at low speed than propellers**
- Turbine mixers are satisfactory with mobile liquids, and because of the greater shearing effect, they can deal with more viscous liquids than propeller mixer

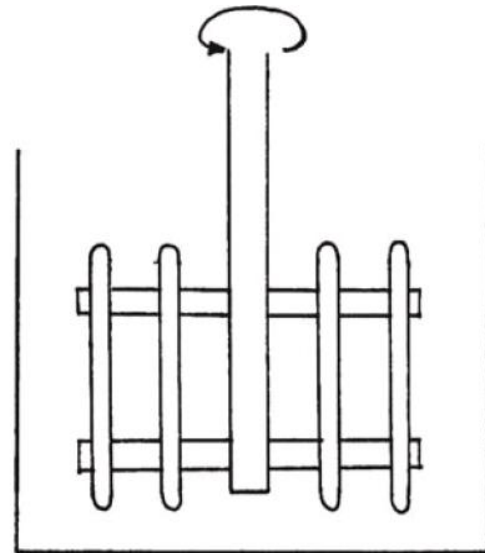
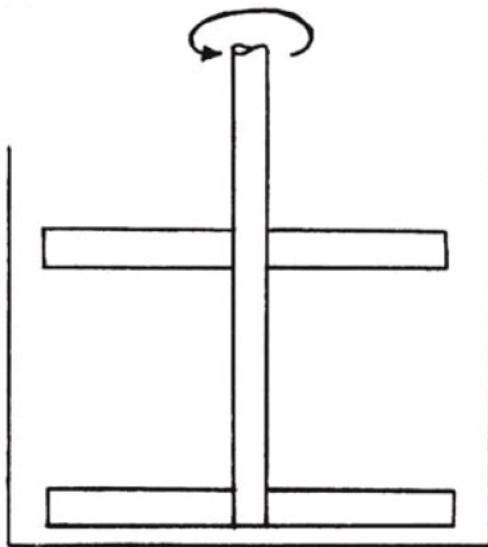


Liquid mixing...

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□ *Paddle mixers*

- Here a number of paddles are used as impeller which consist flat blades attached to shaft and rotate at a low speed of 100 rpm



Solid (powder)mixing

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- The character of powder mixing operation is quite d/t from that of liquids:
 - Here two or more than two solid substances are mixed
 - The objective is to produce a bulk mixture which is then divided into small doses

Solid mixers

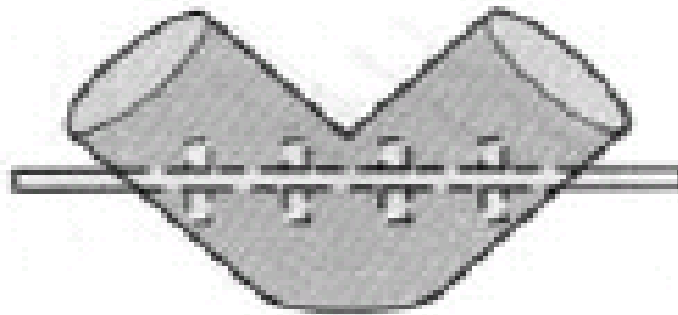
- The equipments used for mixing of powders include:
 - Pestle and mortar
 - Spatula
 - Tumbler mixers
 - Agitation mixers
- } **Small scale**
- } **Large scale**

Solid mixing...

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A) Tumbler mixer

- It consists of metallic container which is rotated on its horizontal axis by means of motor
- The most popular mixer is the twin shell blender 'V' shape

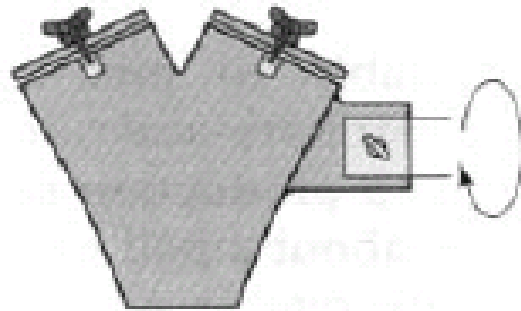


Twin shell (V) mixer
with agitator bar

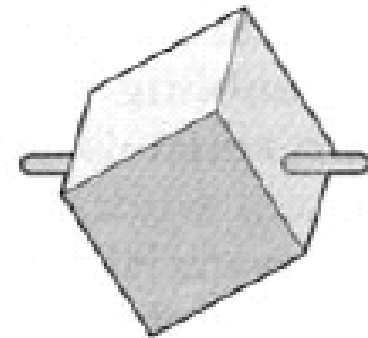
Solid mixing...

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- This is rotated so that the material is alternately collected in the bottom of the V and the split into two portions when the V is inverted
- To improve the efficiency of mixing, a bar fitted with blades which rotates on the opposite direction to that of blender may be attached
- Various other shapes include: cubical, Y shaped and cylindrical



Y-cone mixer



Rotating cube

Solid mixing...

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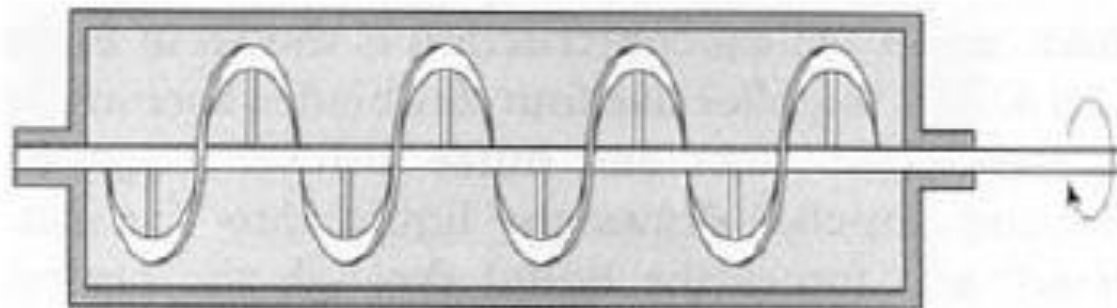
- The efficiency of tumbling mixers are highly dependent on the *speed of rotation*
- Rotation that is **too slow** doesn't produce the desired intense **tumbling** or cascading motion
- Rotation that is **too rapid** tends to produce centrifugal force sufficient to hold the powder to the sides of the mixer and thereby reduce efficiency
- The optimum rate of rotation depends on *the size and shape of the tumbler* and also *on the type of material being mixed*.

Solid mixing...

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B) Agitator mixer

- Agitator mixers for powders can take similar form to paddle mixers for liquids, but their efficiency is low
- Here the container holding the material is fixed and the mixing is done by means of moving screws, paddles or blades
- **Mixing is effected by means of helical blades**



Mixing of semi-solids

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- The mechanisms involved in mixing semisolids depends on the character of the material, which may show considerable variation.

Mixers for Semi-solids

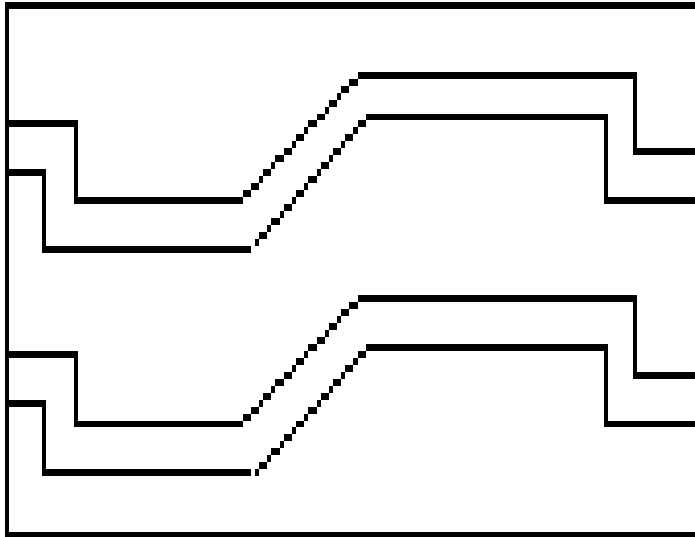
A) Agitator mixers

- Works on the same principle as that used for liquids and powders
- Mixers deigned are of heavier construction to handle materials of heavy consistency
- The agitator arms are designated to give a pulling and kneading action and the shape and movement is such that material is cleared from all sides and corners of the mixing vessel

Mixing of semi-solids...

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- The commonly used one is **sigma arm mixer**
- Used for mixing of **ointments** and **pastes**



Sigma mixer



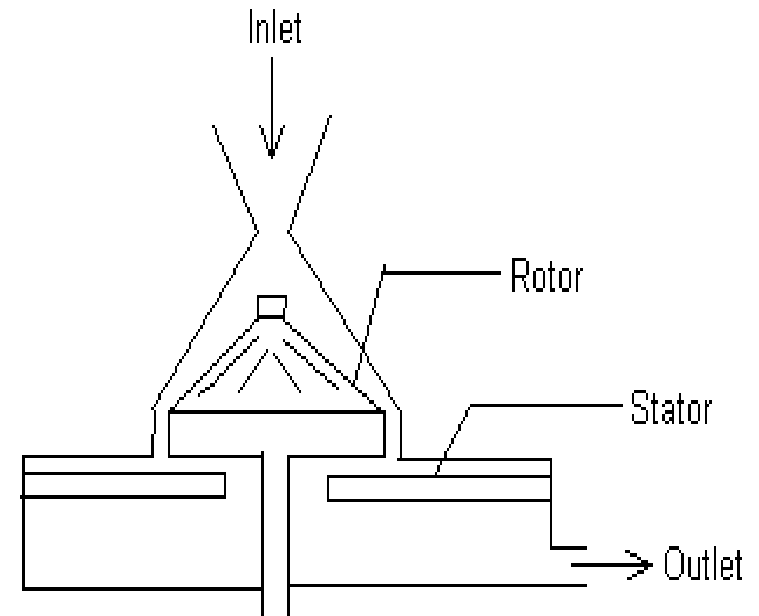
Planetary mixer

Mixing of semi-solids...

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B) Shear mixers

- Machines designed for size reduction can be used for mixing
E.g. roller mill, colloidal mill

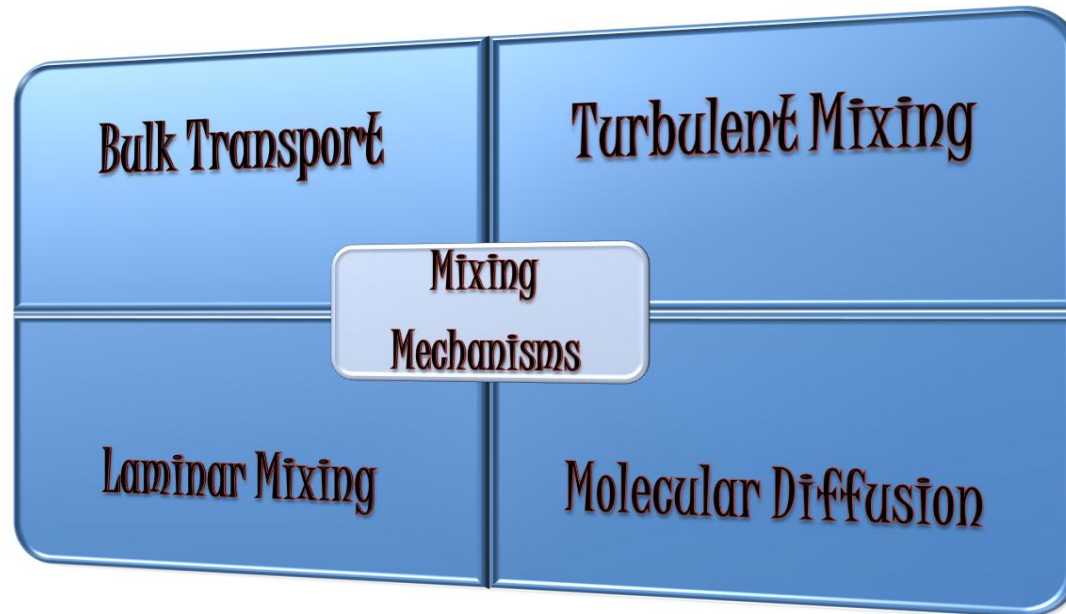


Mixing Mechanisms

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Mixing mechanisms for fluids

- Mixing mechanisms for fluids fall essentially into four categories:



Mixing Mechanisms

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I) Bulk transport

- The movement of a relatively large portion of the material being mixed from one location in the system to another
- A simple circulation of material in a mixer, however, doesn't necessarily result in efficient mixing
- For bulk transport to be effective, it must result in a rearrangement of the various portions to be mixed. *This is usually accomplished by means of paddles or other devices*

Mixing Mechanisms...

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II) Turbulent Mixing

- It is a result of turbulent fluid flow which is characterized by random fluctuation of the fluid velocity at any given point within the system

III) Laminar Mixing

- Laminar flow of fluids which is the flow of fluids without lateral mixing is employed

IV) Molecular diffusion

- Here the diffusion mechanism which is the movement of molecules from area of high concentration to an area of lower concentration is employed

Mixing Mechanisms...

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Mixing mechanisms in solids

I) Convective Mixing

- Here a relatively large mass of material is made to move from one part of the powder bed to another
- Convective mixing can be done by an inversion of the powder bed by means of blades or resolving screws.

II) Shear Mixing

- Here shear forces are created within the mass of the material by using agitator or blast of air.

III) Diffusive Mixing

- Here mixing occurs by diffusion of the powder particles through each other.

Segregation

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- ❖ Segregation happens during or after mixing due to **handling** or **pouring**.

A) Percolation segregation

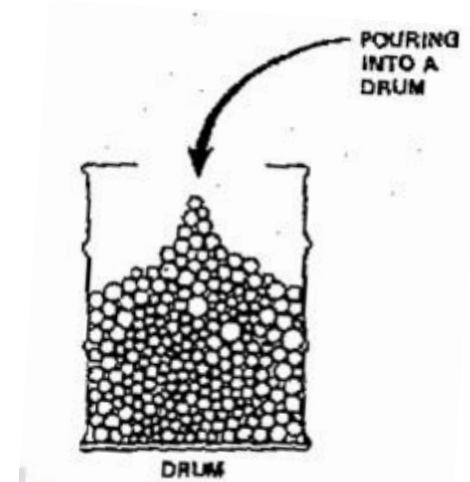
- ❖ The movement of particles through the voids in the powder bed.
- ❖ It will occur in a static bed if the percolating particles are so small that they can fall into the void spaces between the larger particles.
- ❖ Pouring of powders would also lead to segregation either by **rolling** or **dusting**.

Segregation...

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B) Segregation by rolling

- ❖ During pouring, a powder pile is formed and gradually built up.
- ❖ During this pile formation, the larger particles, because of their mass, have tendency to roll down outside the powder pile.
- ❖ On the other hand, fine particles have tendency to stick to the top and concentrate in the middle of the pile.

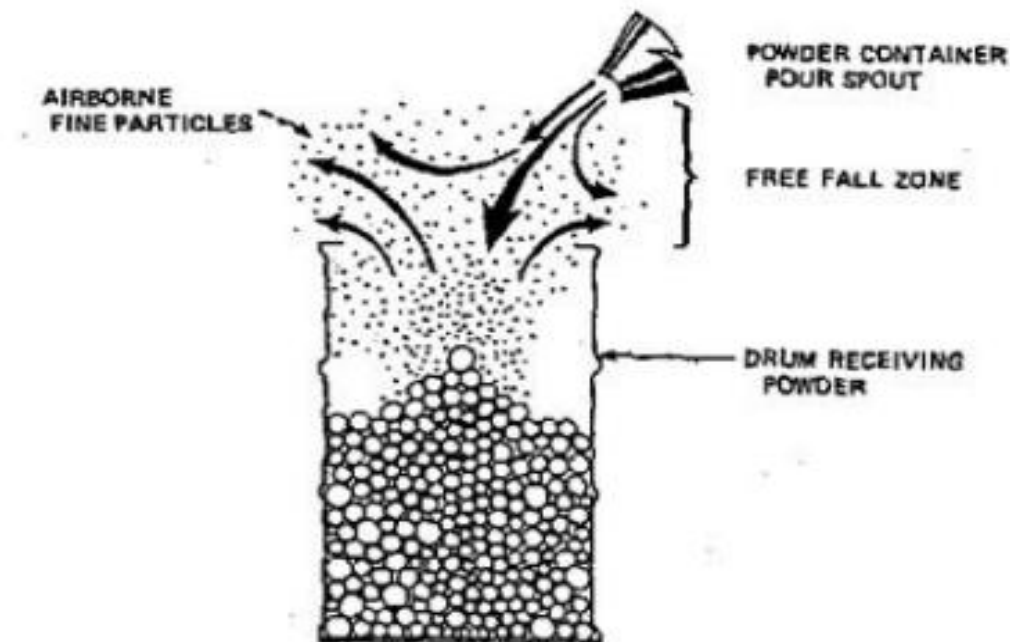


Segregation...

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C) Segregation by dusting

- ❖ In this case the fine particles become air-borne and separate from the bulk of the powder.



Application of Mixing

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- ❑ To make mixtures of uniform composition
- ❑ To mix two miscible liquids
- ❑ To mix two immiscible liquids
- ❑ To dissolve a solid into a vehicle
- ❑ To mix an insoluble solid into a vehicle
- ❑ For mixing of drugs with the base to produce ointments
- ❑ To mix two solid ingredients to make powders